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SNITCH

E-Commerce Clothing Platform

MCA Semester 3 Mini Project
Session: 2026-27

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Project Introduction

- SNITCH is a clothing-focused e-commerce web application developed to provide a simple and modern online shopping experience.
- It allows customers to browse and purchase clothing products, while providing separate modules for sellers and administrators to manage products, orders, users, and other platform activities.
- The project mainly focuses on creating a user-friendly interface that is visually appealing and suitable for modern users, particularly the Gen Z audience.

Problem Statement

- Existing e-commerce platforms offer multiple product categories, making the clothing-shopping experience less focused.
- Users interested specifically in clothing face a cluttered, general-purpose browsing experience.
- There is a need for a dedicated clothing platform with a modern and visually appealing interface Current platforms lack a Gen Z-oriented design language for fashion-focused shopping SNITCH aims to solve this by providing a focused, clothing-only marketplace with dedicated seller and admin portals.

Existing System

- General-purpose e-commerce platforms dominate the market (Amazon, Flipkart, etc.)
- These platforms serve multiple product categories — electronics, groceries, clothing, etc.
- Clothing products are mixed with thousands of other categories, No dedicated clothing-first shopping experience, Seller tools are designed for all product types, not optimized for clothing sellers, Interface design targets a broad audience, not specifically fashion-conscious Gen Z users.

Proposed System



- SNITCH is Clothing-focused e-commerce web application.
- Modern yet simple and Gen Z-oriented UI.
- Dedicated customer shopping experience.
- Dedicated seller portal for clothing products.
- Admin monitoring and management panel.
- Streamlined clothing purchase flow.

Objectives and Scope

OBJECTIVES

- Develop a dedicated e-commerce platform for clothing and fashion.
- Provide a simple, modern and Gen Z-oriented user interface.
- Allow customers to browse, search, filter, Wishlist, cart and purchase.
- Enable sellers to add, manage and sell clothing products.
- Provide administrators tools to manage users, sellers, products and orders.
- Improve overall shopping experience and customer satisfaction.

SCOPE

- Core web-based shopping experience for clothing
- Customer registration, browsing, searching, ordering
- Seller product management and order fulfillment
- Admin platform monitoring and management

Excluded from current version:

- Dedicated mobile application
- Advanced AI-based personalization
- AI-based product suggestions

Literature Survey

Area	Source	Relevance
Frontend Development	React.js Documentation	Component-based UI architecture for building interactive user interfaces
Backend Development	Node.js & Express.js Documentation	Server-side runtime and framework for building RESTful APIs
Database Management	MongoDB Documentation	NoSQL document database for flexible data storage
Version Control	Git Documentation	Source code management and collaboration

Technology Stack

Frontend
React.js

Backend
Node.js, Express.js

Database
MongoDB

Development IDE
Visual Studio Code

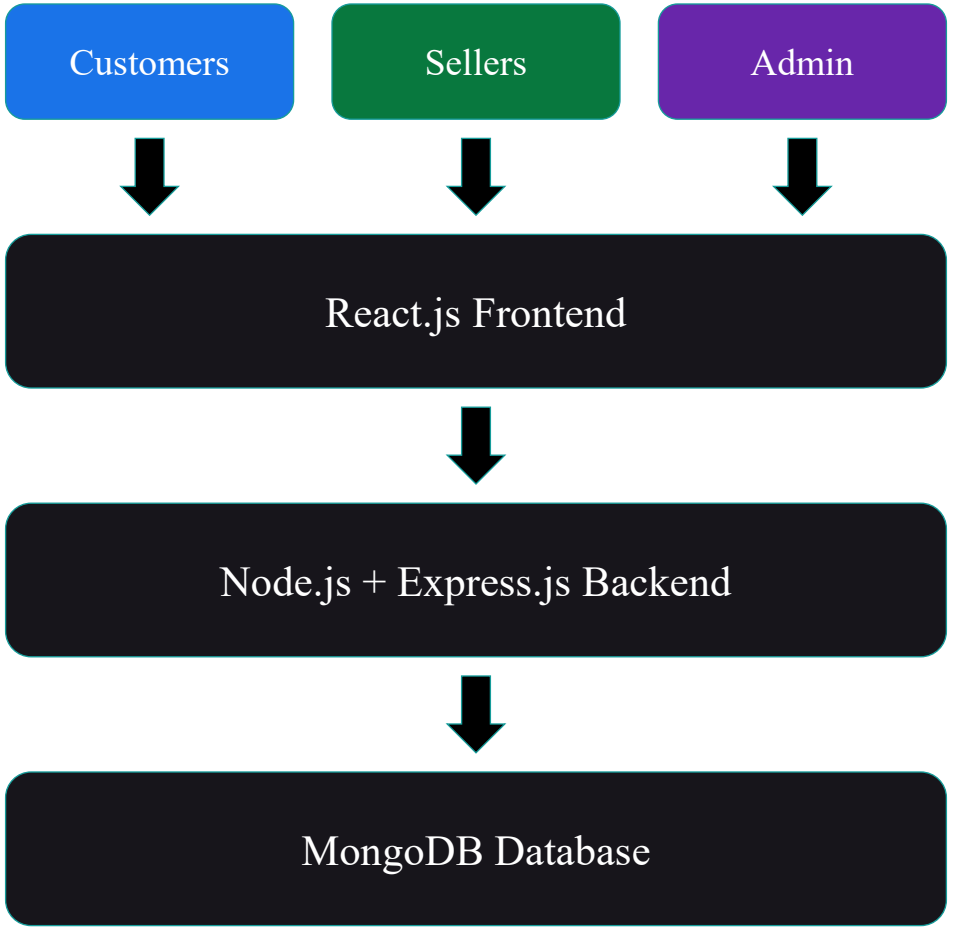
Environment
Modern Web Browser
Windows 10/11

Hardware Requirements:
Modern processor
Minimum 4 GB RAM
Stable internet connection

Payment Interface
Razorpay Payment Gateway

Version Control & Tools
Git, GitHub, Postman for API Testing

System Architecture & Methodology



Requirement Analysis

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System Design

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Module Planning & Wireframes

↓

Development

↓

Testing

↓

Implementation & Deployment

Modules and Key Features

MODULE 1 — CUSTOMER

Landing Page with Featured Categories
User Registration & Login
Product Search with Filter & Sort
Product Details with Size/Color Selection
Wishlist Management
Shopping Cart with Coupon Code
Checkout with Address & Payment (UPI/Card/COD)
Order Tracking & History

MODULE 2 — SELLER

Seller Registration & Login
Seller Dashboard with Sales Overview
Add New Product (Multi-step Wizard)
My Products with Stock Management
Order Management & Fulfillment
Sales Reports with Charts
Store Profile & Business Settings

MODULE 3 — ADMIN

Admin Login
Platform Dashboard with Key Metrics
Customer Management (View/Block)
Seller Management (Approve/Suspend)
Product Monitoring (View/Hide/Delete)
Order Management (View/Refund/Cancel)
Reports & Analytics
Admin Settings

Implementation, Results & Testing

IMPLEMENTATION

Complete wireframe design for all three modules
Customer flow: 12 screens across 3 pages
Seller flow: 10 screens across 3 pages
Admin flow: 8 screens across 2 pages

RESULTS

Full end-to-end user journey mapped for Customer, Seller and Admin
Realistic e-commerce workflows demonstrated through wireframes
Consistent design language across all modules

TESTING

Wireframe review and layout validation
Cross-browser rendering verification
Print layout testing (A4 Landscape)
Component independence verification

Conclusion & Future Scope

CONCLUSION

SNITCH provides a focused, clothing-only e-commerce experience
The wireframe design covers complete user journeys for all three modules
The platform is designed with a modern, Gen Z-oriented approach
The project establishes a strong foundation for full-stack development

FUTURE SCOPE

Full-stack implementation using React.js, Node.js and MongoDB
Dedicated mobile application
AI-based personalization and product recommendations
Payment gateway integration
Real-time order tracking and notifications

Reporting - 2

UML Diagrams

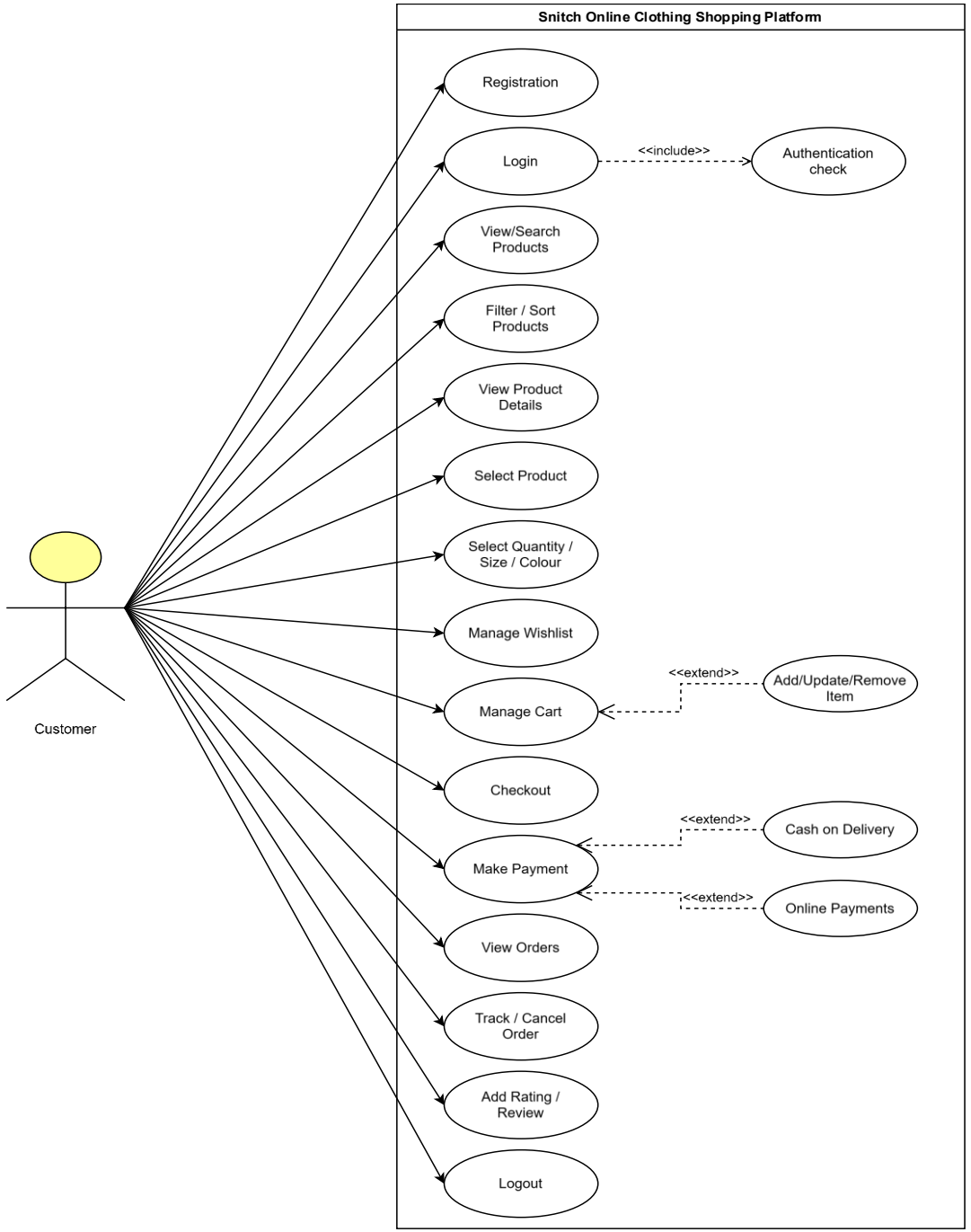
Admin Use Case Diagram :



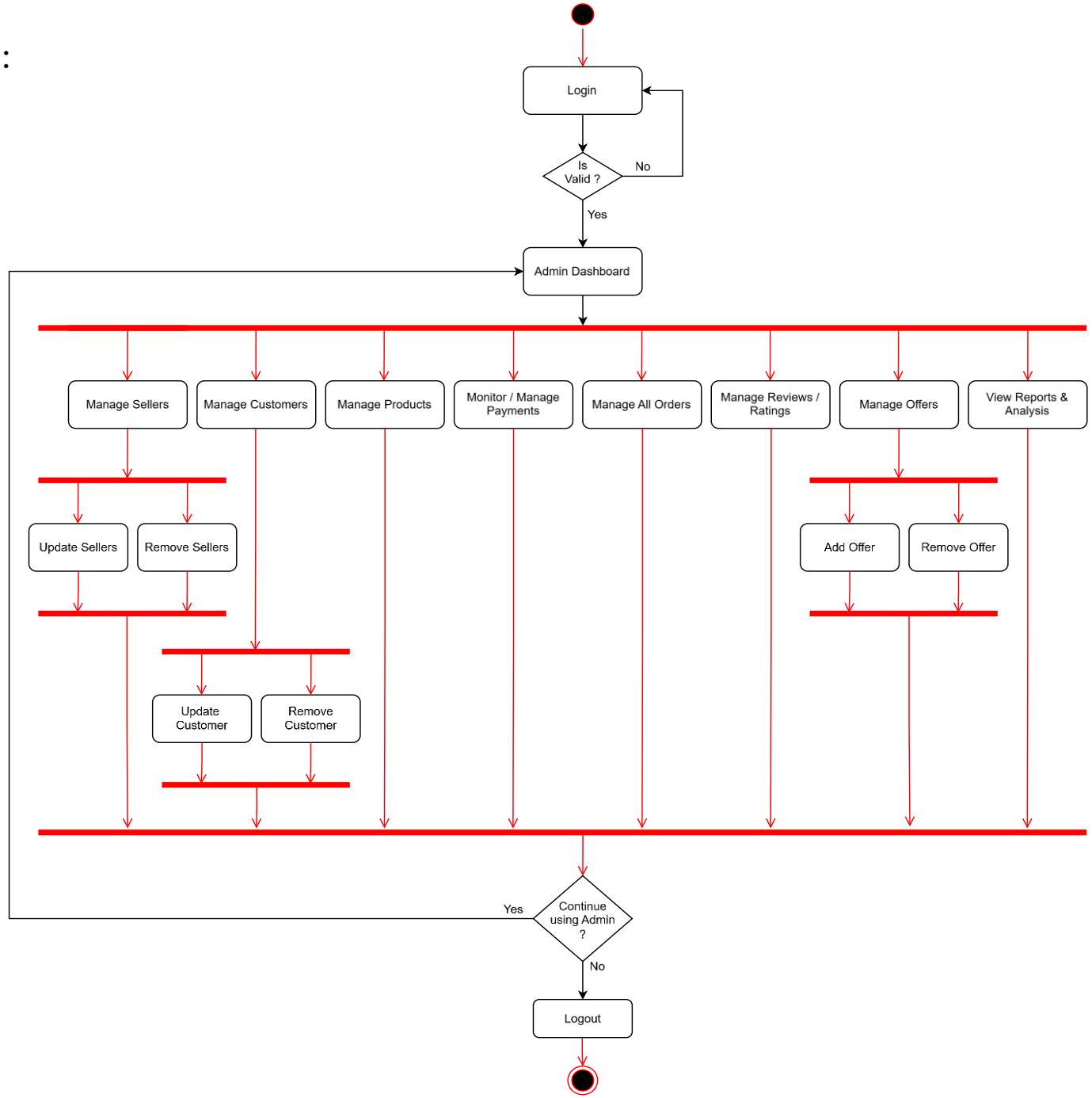
Seller Use Case Diagram :



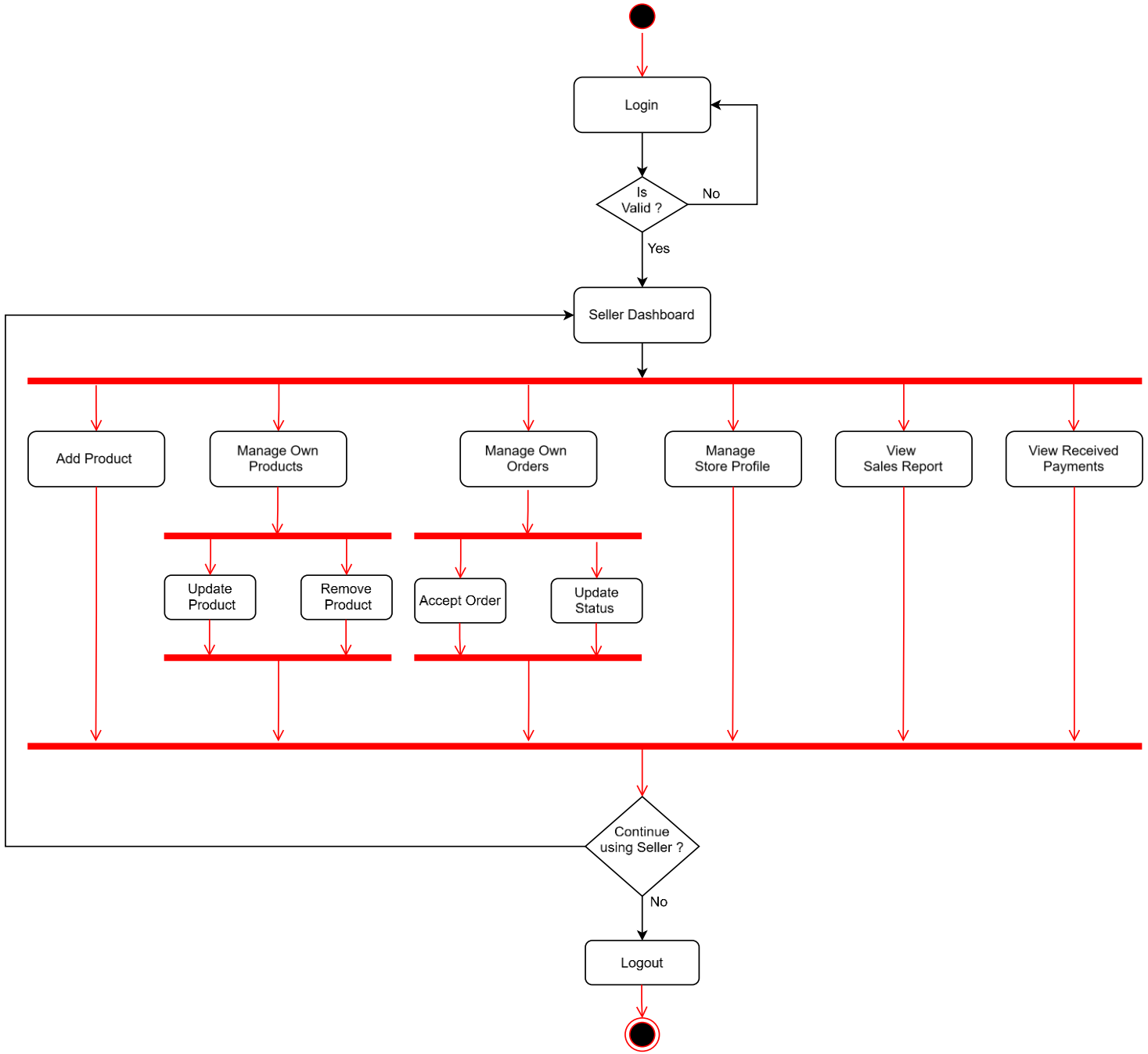
Customer Use Case Diagram :



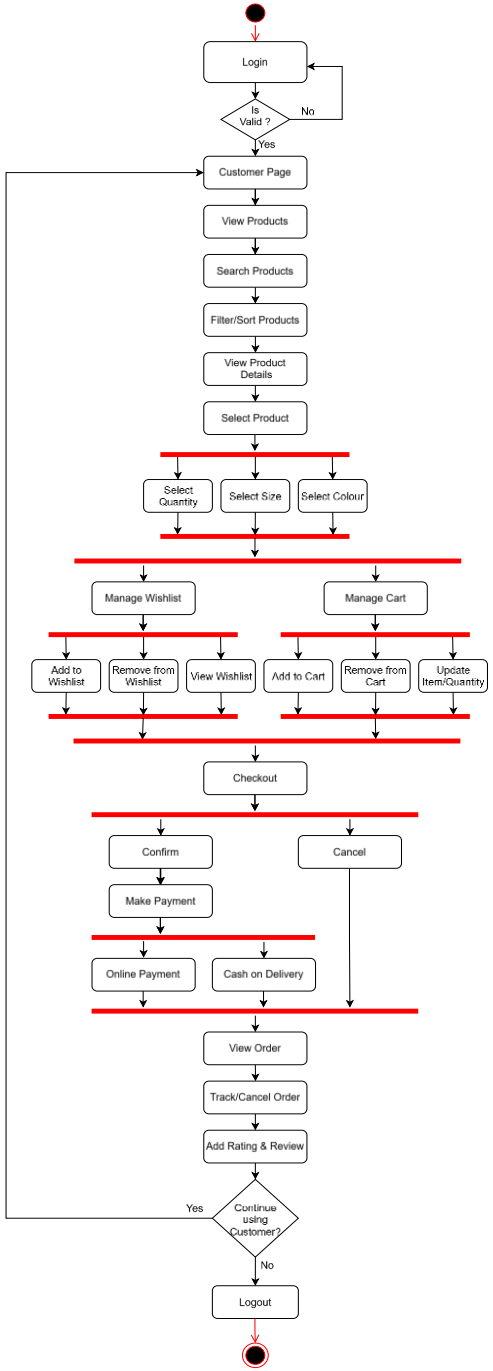
Admin Activity Diagram :



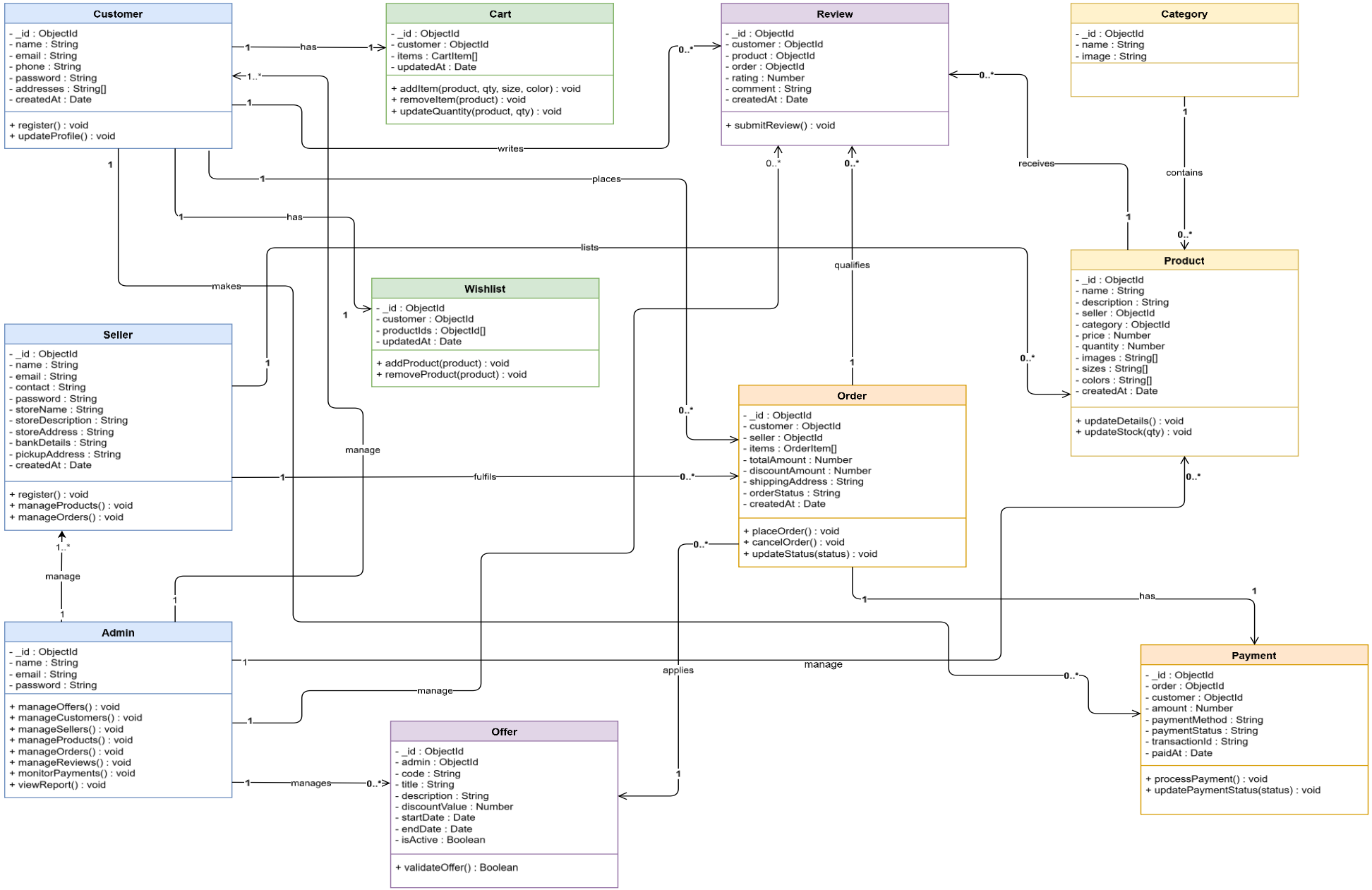
Seller Activity Diagram :



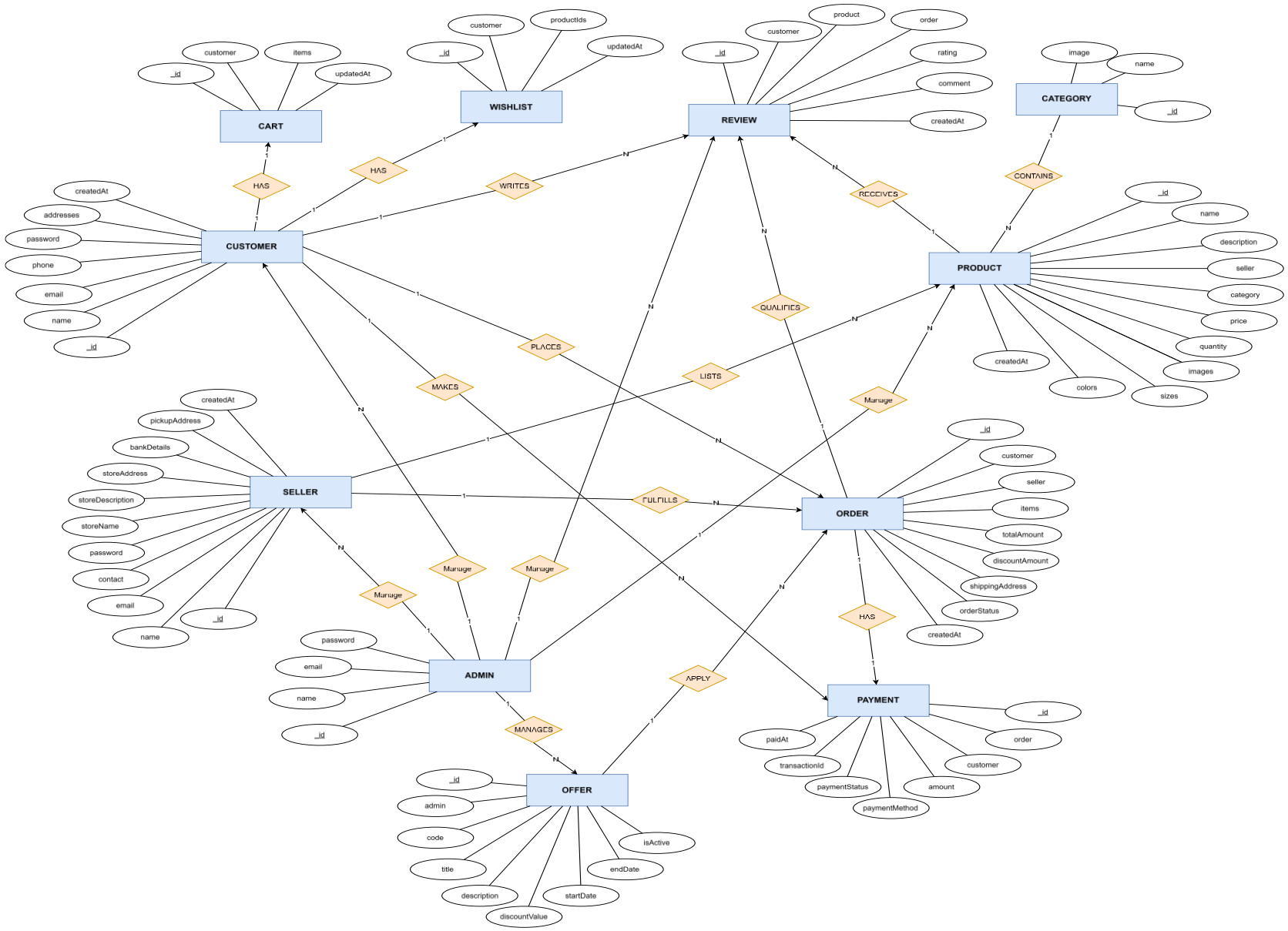
Customer Activity Diagram :



Class Diagram :



Entity Relationship Diagram :



Data Dictionary

1. Admin Table :

No	Field Name	Data type	Constraints	Description
1.	_id	ObjectId	Primary document identifier; automatically generated	Uniquely identifies each administrator document.
2.	name	String	Required	Stores the administrator's name.
3.	email	String	Required, Unique	Stores the administrator's email address for login and communication.
4.	password	String	Required	Stores the administrator's hashed password.

2. Seller Table :

No	Field Name	Data type	Constraints	Description
1.	_id	ObjectId	Automatically generated; document identifier	Uniquely identifies each seller document.
2.	Name	String	Required	Stores the seller's name.
3.	email	String	Required, Unique	Stores the seller's email address for login and communication.
4.	conatct	String	Required	Stores the seller's contact number.
5.	password	String	Required	Stores the seller's hashed password.
6.	storeName	String	Required	Stores the name of the seller’s clothing store.
7.	storeDescription	String	Required	Describes the seller's store and its offerings.
8.	storeAddress	String	Required	Store the address of the seller’s store.
9.	bankDetails	String/Object	Required	Stores the bank details needed for seller payment settlements.
10.	pickupAddress	String	Required	Stores the address from which orders can be picked up for delivery.
11.	createdAt	Date	Automatically assigned	Records when the seller account was created.

3. Customer Table :

No	Field Name	Data type	Constraints	Description
1.	_id	ObjectId	Automatically generated; document identifier	Uniquely identifies each customer document.
2.	name	String	Required	Stores the customer's full name.
3.	email	String	Required, Unique	Stores the customer's email address for login and communication.
4.	conatct	String	Required	Stores the customer’s contact number.
5.	password	String	Required	Stores the customer’s hashed password
6.	Addresses	Array of Objects	Optional	Stores the customer’s saved delivery addresses.
7.	createdAt	Date	Automatically assigned	Records when the customer account was created.

4. Product Table :

No	Field Name	Data type	Constraints	Description
1.	_id	ObjectId	Automatically generated; document identifier	Uniquely identifies each product.
2.	Name	String	Required	Name of the clothing product.
3.	description	String	Required	Detailed description of the product.
4.	seller	ObjectId	Required; references Seller	Identifies the seller who lists the product.
5.	category	ObjectId	Required; references Category	Identifies the category to which the product belongs
6.	price	Number	Required; Non negative	Selling price of the product.
7.	quantity	Number	Required; Integer greater than or equal to 0	Available stock of the product.
8.	images	Array of String	Required	Store images UPLs for the products.
9.	sizes	Array of String	Required	Stores available clothing sizes, such as S, M, L, XL
10.	colors	Array of String	Required	Stores available product colors.
11.	createdAt	Date	Automatically assigned	Records when the product account was created.

5. Category Table :

No.	Field name	Data type	Constraints	Description
1.	_id	ObjectId	Auto-generated; document identifier	Uniquely identifies each category.
2.	name	String	Required; unique	Name of the product category, such as Men, Women, or Accessories.
3.	image	String	Optional	Stores the URL or path of the category image.

6. Cart Table :

No	Field name	Data type	Constraints	Description
1.	_id	ObjectId	Auto-generated; document identifier	Uniquely identifies each cart.
2.	customer	ObjectId	Required; references Customer; unique	Identifies the customer who owns the cart. Ensures one cart per customer.
3.	items	Array of Objects	Optional; each item contains product details	Stores the products added to the cart, along with their quantities and selected options.
4.	updatedAt	Date	Automatically updated when the cart changes	Records the date and time of the cart's last update.

7. Wishlist Table :

No	Field name	Data type	Constraints	Description
1.	_id	ObjectId	Auto-generated; document identifier	Uniquely identifies each wishlist.
2.	customer	ObjectId	Required; references Customer; unique	Identifies the customer who owns the wishlist. Ensures one wishlist per customer.
3.	productIds	Array of ObjectIds	Optional; each ID references Product	Stores the IDs of products saved by the customer for future purchase.
4.	updatedAt	Date	Automatically updated when the wishlist changes	Records the date and time of the wishlist's last update.

8. Order Table :

No	Field name	Data type	Constraints	Description
1.	_id	ObjectId	Auto-generated; document identifier	Uniquely identifies each order.
2.	customer	ObjectId	Required; references Customer	Identifies the customer who placed the order.
3.	seller	ObjectId	Required; references Seller	Identifies the seller responsible for fulfilling the order.
4.	items	Array of Objects	Required; must contain at least one item	Stores the products included in the order and their purchase details.
5.	totalAmount	Number	Required; non-negative	Stores the total amount payable for the order.
6.	discountAmount	Number	Optional; non-negative; defaults to 0	Stores the discount applied to the order.
7.	shippingAddress	String	Required	Stores the delivery address for the order.
8.	orderStatus	String	Required; restricted to predefined values	Tracks the order's current status, such as Pending, Confirmed, Shipped, Delivered, or Cancelled.
9.	createdAt	Date	Automatically assigned during creation	Records when the order was placed.

9. Payment Table :

No	Field name	Data type	Constraints	Description
1.	_id	ObjectId	Auto-generated; document identifier	Uniquely identifies each payment record.
2.	order	ObjectId	Required; references Order; unique	Identifies the order associated with the payment. Supports the designed one-to-one Order–Payment relationship.
3.	customer	ObjectId	Required; references Customer	Identifies the customer who made the payment.
4.	amount	Number	Required; non-negative	Stores the amount paid or to be paid for the order.
5.	paymentMethod	String	Required; restricted to predefined values	Specifies the payment method, such as UPI, Card, or COD (Cash on Delivery).
6.	paymentStatus	String	Required; restricted to predefined values	Tracks the payment status, such as Pending, Paid, Failed, or Refunded.
7.	transactionId	String	Optional	Stores the transaction identifier returned by the payment provider, when applicable.
8.	paidAt	Date	Optional	Records the date and time when the payment was completed.

10. Review Table :

No	Field name	Data type	Constraints	Description
1.	_id	ObjectId	Auto-generated; document identifier	Uniquely identifies each review.
2.	customer	ObjectId	Required; references Customer	Identifies the customer who wrote the review.
3.	product	ObjectId	Required; references Product	Identifies the product being reviewed.
4.	order	ObjectId	Required; references Order	Identifies the order associated with the reviewed product.
5.	rating	Number	Required; integer between 1 and 5	Stores the customer's rating for the product.
6.	comment	String	Optional	Stores the customer's written feedback about the product.
7.	createdAt	Date	Automatically assigned during creation	Records when the review was submitted.

11. Offer Table :

No	Field name	Data type	Constraints	Description
1.	_id	ObjectId	Auto-generated; document identifier	Uniquely identifies each offer.
2.	admin	ObjectId	Required; references Admin	Identifies the administrator who manages the offer.
3.	code	String	Required; unique	Stores the offer or discount code customers can use.
4.	title	String	Required	Stores the title or name of the offer.
5.	description	String	Optional	Provides details about the offer and its terms.
6.	discountValue	Number	Required; non-negative	Stores the discount value provided by the offer.
7.	startDate	Date	Required	Specifies when the offer becomes valid.
8.	endDate	Date	Required; must be later than startDate	Specifies when the offer expires.
9.	isActive	Boolean	Defaults to true	Indicates whether the offer is enabled.

Thank you